

Predicates and Quantifiers

What Propositional Logic Cannot Do

- *We saw that some declarative sentences are not propositions without specifying the value of 'indeterminates'*

“ $x + 2$ is an even number”

“If $x + 1 > 0$, then $x > 0$ ”

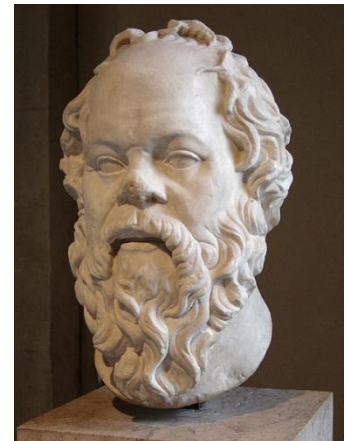
“ x drives a lambo”

- *Some valid arguments cannot be expressed with all our machinery of tautologies, equivalences, and rules of inference*

Every man is mortal.

Socrates is a man.

$\therefore$ Socrates is mortal



Predicates

- Sentences like ' x is greater than 3' or 'person x has a brother' are not true or false unless the variable is assigned some particular value.
- The sentence ' x is greater than 3' consists of 2 parts.
 - The first part, x , is called the **variable** or the **subject** of the sentence.
 - The second part – the **predicate** 'is greater than 3' – refers to a **property the subject can have**.
- Sentences that have such structure are called **predicates**
- We write $P(x)$ to denote a predicate with variable x

Unary, Binary, and so on

`x is greater than 3'
`x is my brother'
`x is a human being'

} ***unary** predicates contain only 1 variable*
 $P(x)$

`x is greater than y'
`x is the mother of y'
`car x has colour y'

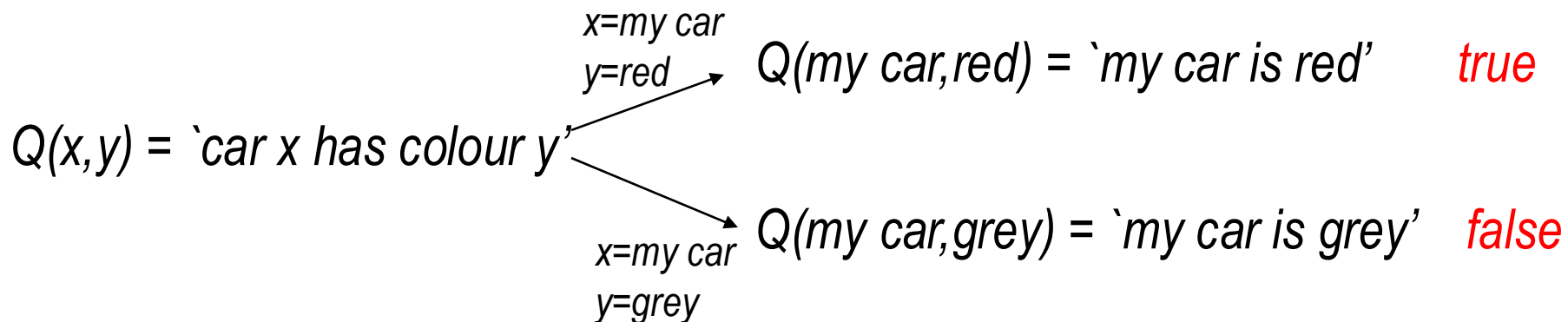
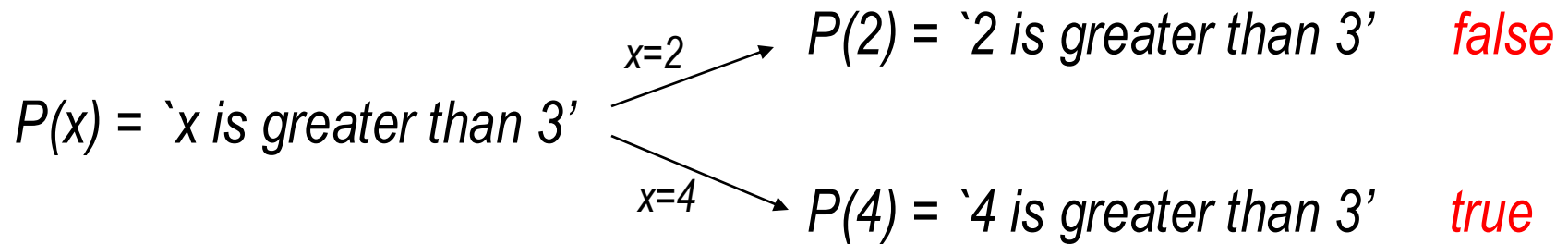
} ***binary** predicates contain 2 variables*
 $Q(x,y)$

`x divides y + z'
`x sits between y and z'
`x is a son of y and z'

} ***ternary** predicates contain 3 variables*
 $R(x,y,z)$

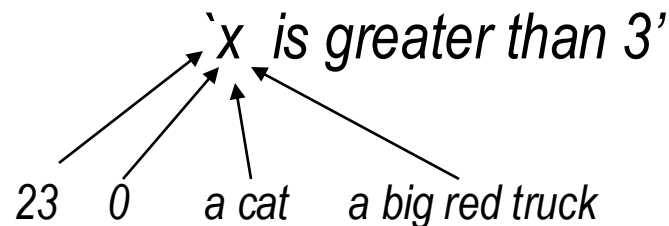
Assigning a Value

- When a variable is assigned a value, the predicate turns into a *proposition*, whose truth value can be evaluated.



Domain

- *We cannot assign a variable of a predicate ANY value. We need to obtain a meaningful statement!*



- *Every variable of a predicate is associated with a **domain** or **domain of discourse**, and its values are taken from this domain*

'x is greater than 3'

x is a number

'x is my brother'

x is a human

'x is an animal'

x is a ???

'car x has colour y'

x is a car
y is a colour

In CS lingo:
variables are **typed**

Example

● $R(x,y,z) = "x + y = z"$

What is the truth value of:

- $R(1,2,3)$
- $R(10,10,10)$
- $R(\text{monkey}, 2, 3)$

*What value of z makes $R(x,y,z)$ true **given x, y** ?*

Quantification

- Consider the statement "for any numbers x, y , $R(x,y,x+y)$ is true"
- This is a *(universally) quantified* statement
- Quantifiers are ways of obtaining propositions from predicates *without giving specific values*
- Examples:

`Every man is mortal'

`There is x such that x is greater than 3'

`There is a person who is my father'

`For any x , $x^2 \geq 0$ '

quantification

Universal Quantifiers

- *Abbreviates constructions like*
 - For all ...*
 - For any ...*
 - Every ...*
 - Each ...*
- *Asserts that a predicate is true for all values in the domain*
 - 'Every man is mortal'*
 - 'All lions are lazy'*
 - 'For any x , $x^2 \geq 0$ '*
- *Notation: $\forall$*
- *$\forall x$ $P(x)$ is true if and only if $P(a)$ is true **for every a in the domain***

Universal Quantifiers (cntd)

'For any x , $x^2 \geq 0$ ' *true!*

'Every car is grey' *false! my car is not grey*

● $\forall x P(x)$ is false if and only if *there is* at least one value a from the universe such that $P(a)$ is false

● Such a value a is called a *counterexample*

Existential Quantifiers

- *Abbreviates constructions like*
 - For some ...*
 - For at least one ...*
 - There is ...*
 - There exists ...*
- *Asserts that a predicate is true for at least one value in the domain*
 - 'There is a living king'*
 - 'Some people are bad'*
 - 'There is x such that $x^2 \geq 10$ '*
- *Notation: $\exists$*
- $\exists x$ $P(x)$ *is true if there is some value a in the domain such that $P(a)$ is true*

Existential Quantifiers (cntd)

'There is a grey car' true! I've seen them

'For some x , $x^2 < 0$ ' false!

- *$\exists x P(x)$ is false if and only if for all a from the universe $P(a)$ is false*
- *Disproving an existential statement is difficult!*

Quantifiers and Negations

● Summarizing

	<i>true</i>	<i>false</i>
$\forall x P(x)$	<i>For every value a from the universe $P(a)$ is true</i>	<i>There is a counterexample – a value a from the universe such that $P(a)$ is false</i>
$\exists x P(x)$	<i>There is a value a from the universe such that $P(a)$ is true</i>	<i>For all values a from the universe $P(a)$ is false</i>

● Observe that

$\neg \forall x P(x)$ is true if and only if $\exists x \neg P(x)$ is true

$\neg \exists x P(x)$ is true if and only if $\forall x \neg P(x)$ is true

DeMorgan laws

Example

- *What is the negation of each of the following statements?*

Statement

Negation

All lions are lazy

There is an active lion

Everyone has two legs

*There is a person who does not
have two legs*

Some people like coffee

All people hate coffee

*There is a lady in
one of these rooms*

There is a tiger in every room

Multiple quantifiers

- *Often predicates have more than one variable. In this case we need more than one quantifier.*

$P(x,y)$ = *“car x has colour y”*

$\forall x \forall y P(x,y)$

“every car is painted all colours

$\exists x \exists y P(x,y)$

“there is a car that is painted some colour

$\forall x \exists y P(x,y)$

“every car is painted some colour

$\exists x \forall y P(x,y)$

“there is a car that is painted all colours



Partial quantification

- A statement containing multiple variables may be quantified in only **some** variables.

$P(x,y)$ = “car x has colour y ”

$\forall y P(x,y)$

“car x is painted all colours

$\exists y P(x,y)$

“car x is painted some colour

Variable y is
bound by the
quantifier

Variable x is
free (not bound
by a quantifier)

Free and bound variables

- Quantifiers bind tightly, and the part of the formula it applies to is called its **scope**

$$\forall y \text{ (.....) } \text{.....}$$

- Occurrences of the quantified variable inside the scope are bound by the quantifier, and are free outside the scope

$$\forall y \text{ } Q(x,y) \vee \forall y(x < y) \vee R(y)$$

bound
bound
free

- Partial quantification gives a *new predicate*

● Let $P(x,y)$ mean “lion x likes y ”

$R(x) = \forall y P(x,y)$ “lion x likes everything” $R(x)$ is always false
(say it using quantifiers: $\forall x \neg R(x)$ or $\forall x \neg(\forall y P(x,y))$)

May not be the same y for Bob and Gob!

Practice

Exercises from the Book:

7th edition: No. 1, 5, 7, 10, 20 (page 53 – 54)

8th edition: No. 1, 5, 7, 10, 20 (page 56 – 57)